XPERTSEDGE TECHNOLOGIES

AWS & DevOps Professional Program

25-Module Hands-On Learning Roadmap

Build practical cloud and DevOps skills through guided labs, automation workflows and portfolio-ready projects across AWS, Linux, CI/CD, containers, infrastructure as code and Kubernetes.

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25 Modules
Structured Learning

AWS

AWS + Linux
Cloud Foundation

Hands-On Labs
Service-Based Practice

CI

CI/CD + Automation
DevOps Toolchain

Capstone Projects
Portfolio Readiness

HR

Career Support
Trainer Guidance

AWS **Learn AWS • Automate Delivery • Build Real Projects**

ABOUT US

About XpertsEdge Technologies

XpertsEdge Technologies is a career-focused IT training institute dedicated to delivering practical, job-oriented training for students, freshers, and working professionals. Our curriculum is built around real-time examples, hands-on practice, and project-based learning that mirrors actual workplace environments. We go beyond theory - every session is designed to build confidence, sharpen technical skills, and prepare learners for real job interviews. With personalized trainer guidance, individual doubt support, and structured assessment cycles, we help learners become job-ready by the end of the program.

Why Choose Us?

- Experienced Industry Trainers
- Practical Hands-on Sessions
- Real-Time AWS & DevOps Projects
- Beginner-Friendly Teaching Style
- Individual Doubt Support
- Job-Oriented Syllabus
- Mock Interview Preparation
- Certificate After Completion

✓ **Certificate of Completion awarded to successful students after finishing the program.**



Job-Oriented Training

Focused curriculum designed for cloud and DevOps career readiness.



Hands-On Projects

Apply skills with practical, real-world AWS assignments.



Personalized Coaching

Receive one-on-one guidance from trainers.

COURSE OVERVIEW

Your Complete AWS & DevOps Learning Roadmap

The program starts with cloud concepts and Linux, progresses through core and advanced AWS services, and then covers Git, Jenkins, monitoring, Ansible, Docker, Terraform and Kubernetes. Every stage includes practical tasks and guided projects.

FOUNDATION

Stage 1

Cloud, Linux, IAM, EC2, Storage and VPC

AWS CORE

Stage 2

Scaling, Databases, Monitoring and Serverless

DEVOPS

Stage 3

Git, Jenkins, Observability, Automation and Containers

i **Program outcome: Create AWS infrastructure, automate delivery, operate containers, build monitoring dashboards and confidently explain an end-to-end DevOps project.**



MODULES
01-04

Cloud Foundations, Linux, IAM & EC2

The first stage establishes cloud concepts, command-line confidence, secure AWS access and compute fundamentals. Learners finish by launching and accessing Linux and Windows EC2 instances.

1

Cloud Computing & AWS

- Cloud models and virtualization
- AWS Regions and Availability Zones
- Global infrastructure
- Selecting an AWS Region

2

Linux Basics

- Navigation and file commands
- Permissions and ownership
- Text processing and archives
- Package management

3

AWS IAM

- Users, groups and roles
- Policies and JSON structure
- Access keys and MFA
- Least-privilege access

4

Amazon EC2

- AMI and instance types
- Key pairs and security groups
- Linux and Windows instances
- Application hosting

✓ Hands-on outcome: Configure secure IAM access, launch EC2, connect through SSH and publish a frontend application.

MODULES 05-09

AWS Storage, Networking and Scaling

This stage connects block, object and shared storage with VPC networking. Learners also configure load balancing and Auto Scaling for resilient AWS applications.

1 EBS and S3

- EBS volume types and attachment
- Snapshots and backups
- S3 buckets, objects and ACLs
- Versioning and storage classes

2 EFS and VPC

- EFS file systems and NFS mounting
- CIDR and subnet design
- Route tables and gateways
- NACLs and security groups

3 ELB and Auto Scaling

- Load Balancer types
- Target groups and health checks
- Scaling policies and options
- Monitoring scaling events

✓ Hands-on outcome: Attach storage, build a four-subnet VPC and distribute application traffic across healthy instances.

MODULES 10-14

Messaging, Monitoring, DNS and Databases

This stage covers asynchronous messaging, operational monitoring, DNS and both relational and NoSQL databases for AWS applications.

SQS and SNS

- Queues and messages
- Topics and subscriptions
- Send, receive and delete
- Connect SNS with SQS

CloudWatch and Route 53

- Metrics, logs and alarms
- Dashboards and notifications
- DNS records and routing policies
- Domain integration

RDS and DynamoDB

- RDS engines and instance types
- Users, tables and SQL queries
- DynamoDB keys and indexes
- CRUD operations

Hands-on outcome: Configure CloudWatch alerts, connect a domain, launch an RDS database and build a DynamoDB table with indexes.



MODULES 15-18

Cloud Delivery, Serverless and DevOps Fundamentals

This stage connects content delivery and audit logging with Lambda and CloudFormation, then introduces DevOps culture and version control using Git and GitHub.

1

CloudFront and CloudTrail

- S3 origin and CDN delivery
- Caching and security
- Trails and audit logging
- Store logs in S3

2

Lambda and CloudFormation

- Functions and integrations
- Monitoring and security
- YAML templates
- Infrastructure deployment

3

DevOps, Git and GitHub

- SDLC, Agile and DevOps lifecycle
- Commits, branches and merges
- Pull requests and conflicts
- Git workflows and best practices

✓ Hands-on outcome: Configure CloudFront, review CloudTrail events, deploy a CloudFormation stack and resolve a Git merge conflict.

MODULES 19-21

Jenkins, Splunk, Prometheus and Grafana

This stage builds the delivery and observability toolchain. Learners configure Jenkins automation, centralize logs in Splunk and visualize infrastructure metrics with Prometheus and Grafana.

Jenkins CI/CD

- Installation and job configuration
- Users, roles and plugins
- Controller and agent setup
- GitHub-to-Jenkins pipeline

Splunk Monitoring

- Enterprise server and forwarder
- Indexing, search and SPL
- Alerts and dashboards
- Centralized AWS log collection

Prometheus and Grafana

→ **Prometheus**	Metrics and PromQL
→ **Node Exporter**	Host metrics
→ **Grafana**	Dashboards and alerts
→ **Sharing**	Dashboard collaboration

Hands-on outcome: Build a Jenkins pipeline, forward AWS logs to Splunk and create infrastructure dashboards in Grafana.

MODULES 22-25

Ansible, Docker, Terraform and Kubernetes

The final stage focuses on configuration automation, containerization, infrastructure as code and container orchestration, followed by practical capstone projects.

Ansible and Docker

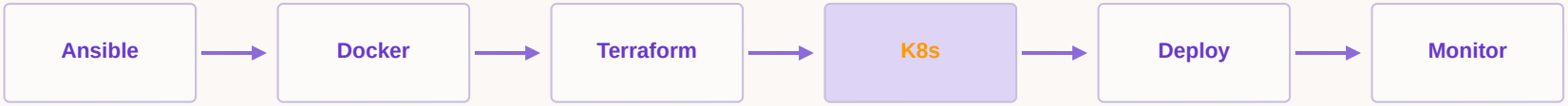
- Inventory and YAML playbooks
- Roles and Ansible Vault
- Images, containers and Dockerfiles
- Push custom images to Docker Hub

Terraform and Kubernetes

- Providers, resources and state
- Init, plan, apply and destroy
- Deployments, services and ingress
- Kubernetes YAML manifests

Capstone Projects

- AWS cost optimization strategy
- Event-driven serverless architecture
- Automated AWS infrastructure
- Kubernetes web application deployment



✓ Capstone outcome: Automate infrastructure, package an application, deploy it to Kubernetes and present the complete delivery workflow.

CAREER SUPPORT

Turn Your Learning into a DevOps Portfolio

Career Preparation

→ **Technical interview practice**

→ **Project explanation practice**

→ **Hands-on assessments**

→ **Resume and profile guidance**

Career Roles You Can Apply For

- DevOps Engineer
- AWS Cloud Engineer
- Cloud Support Engineer
- CI/CD Engineer
- Junior SRE
- Platform Engineer

Build Skills. Complete Projects. Prepare for Interviews.

Complete guided labs, build portfolio projects and practise explaining every workflow with support from experienced trainers.

Tools You Will Master



CI

AWS + Linux

Cloud and system foundation

IaC

Git + Jenkins

Version control and CI/CD

K8s

Docker + Terraform

Containers and infrastructure as code

Kubernetes + Ansible

Orchestration and configuration

OBS

Splunk + Grafana

Monitoring and observability

Course Enquiries **XpertsEdge Technologies**

Contact our training team for batch details

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